

MMRD Nova Distance Calculator

The **MMRD nova distance calculator** estimates the distance to a nova from its rate of decline after maximum light. It is intended to do for novae what the Leavitt's Law distance calculator does for Cepheids: combine a measured light-curve quantity with a published luminosity calibration and the distance modulus.

This plug-in appears in VStar's **Tool** menu as **MMRD nova distance calculator**.

References

The implementation is based primarily on:

- Kantharia, N. G. 2017, *Novae: I. The maximum magnitude relation with decline time (MMRD) and distance*, arXiv:1703.04087.
- Kok, Y. 2010, *Absolute Magnitudes and Distances of Recent Novae*, JAAVSO, 38, 193.

The historical MMRD calibrations are taken from Kok's summary of Schmidt (1957), Cohen (1985), della Valle & Livio (1995), and Downes & Duerbeck (2000), with the sign convention cross-checked against Kantharia (2017).

What the Calculator Does

The maximum magnitude versus rate of decline (MMRD) method estimates the peak absolute magnitude, M_V , from the time taken for a nova to decline by 2 or 3 magnitudes from maximum:

- t_2 : days from maximum to maximum + 2 magnitudes.
- t_3 : days from maximum to maximum + 3 magnitudes.

Once M_V is estimated, the distance is computed from the extinction-corrected distance modulus:

$$D = 10^{0.2(m_V - A_V - M_V + 5)} \text{ pc}$$

where:

- m_V is the peak apparent magnitude,
- A_V is visual extinction,
- D is distance in parsecs.

The result dialog reports distance in kiloparsecs (kpc), which is the unit used in the nova distance literature, and includes light years for the point estimate as a convenience.

If $A_V = 0$, the reported distance should be treated as an upper limit unless extinction is known to be negligible.

Basic Workflow

1. Load nova observations, preferably around the outburst and early decline.
2. Choose **Tool > MMRD nova distance calculator**.
3. Select the observation series to use.
4. Choose a source for t_2 and t_3 :
 - **Exponential model fit (Kok 2010, eq. 10)** for raw visual or V data.
 - **Directly from observations** only for clean or already-smoothed data.
5. Review or edit m_V , t_2 , t_3 , optional σt_2 and σt_3 , and extinction.
6. Choose an MMRD relation. The equation and error source update as you change the drop-down.
7. Click **OK**. The results dialog shows the selected relation in detail and a summary table of **every** MMRD relation applied to the same m_V , t_2 , t_3 , and A_V .
8. Dismiss the results dialog to return to the input dialog. You can then choose a different relation, adjust extinction or decline times, and compare again. Series selection and the exponential fit are **not** repeated. Click **Cancel** on the input dialog to leave the plug-in.

MMRD Inputs

MMRD Relation

Unweighted mean of historical relations (Kok 2010)



MMRD Relation Equation and Error Source

Mean of the historical MMRD relations

Error source: inter-relation scatter

Peak Apparent Mag (mv)

6.88

t2 (days)

4.063138095973

t3 (days)

8.171956925071

sigma t2 (days)

0.303948031226

sigma t3 (days)

0.641307938393

Extinction (Av)

0

or Reddening E(B-V); $A_v = 3.1 E(B-V)$

0

Cancel

OK

MMRD Nova Distance

MMRD Relation						
Unweighted mean of historical relations (Kok 2010)						
Peak Absolute Magnitude (Mv)						
−9.515649 ± 0.294097						
Distance Modulus (mv − Av − Mv)						
16.395649						
Distance (kpc)						
19.016461593856						
Distance (light years)						
61993.664795970515						
Distance Lower Bound (kpc)						
16.607731047101						
Distance Upper Bound (kpc)						
21.774546476277						
Distance Error (kpc)						
−2.408730546755 / +2.758084882421						
Note						
No extinction was applied (Av = 0), so the distances are upper limits.						
All MMRD relations						
Relation	Mv	sigma Mv	D (kpc) ▲	D lower (kpc)	D upper (kpc)	
della Valle & Livio 1995 (arctan, t2)	−8.94	0.01	14.58	14.51	14.65	
Cohen 1985 (linear, t2)	−9.23	0.34	16.69	14.27	19.52	
Inverse-variance weighted mean of historical relations	−9.37	0.29	17.77	15.52	20.35	
Schmidt 1957 (linear, t3)	−9.47	0.09	18.61	17.90	19.36	
Downes & Duerbeck 2000 (linear, t3)	−9.47	0.65	18.64	13.82	25.15	
Kantharia 2017 (linear, t2)	−9.49	0.17	18.78	17.39	20.29	
Unweighted mean of historical relations (Kok 2010)	−9.52	0.29	19.02	16.61	21.77	
Downes & Duerbeck 2000 (arctan, t2)	−9.57	0.02	19.47	19.33	19.62	
Downes & Duerbeck 2000 (linear, t2)	−9.77	0.49	21.35	17.05	26.75	
Downes & Duerbeck 2000 (conditional linear, t3)	−9.82	1.10	21.86	13.16	36.32	
Downes & Duerbeck 2000 (conditional linear, t2)	−9.86	1.16	22.27	13.07	37.94	

Dismiss

Choosing the JD Range

For comparison with Kok (2010), use the same outburst window shown in the paper's figures. For example, Kok's Figure 4 for V4633 Sgr has an x-axis labelled `JD + 2450894.5000` and spans 0 to 60 days, corresponding to:

$$2450894.5 \leq JD \leq 2450954.5$$

Using a substantially longer time range can change the exponential fit, the asymptote, and hence t_2 and t_3 .

Light-Curve Parameter Sources

Exponential Model Fit

This option fits Kok's equation (10):

$$m(t) = P_1 - P_2 e^{-P_3(t-t_0)}$$

The fit is performed from the brightest observation onward. The plug-in then measures t_2 and t_3 as the fitted curve's crossing times at $m_0 + 2$ and $m_0 + 3$, where m_0 is the observed maximum magnitude.

This is the recommended source for raw AID visual data. Raw observations can scatter above and below a smooth decline, so direct crossing detection may find a too-early crossing.

This is the same fit provided by the standalone **Nova Exponential Decline Model** plug-in; see its documentation for the parameters, decline-time derivation, fit metrics, and uncertainty handling.

Directly from Observations

This option finds the brightest observation and linearly interpolates the first crossings at peak + 2 and peak + 3 magnitudes. It is useful for clean, well-sampled, monotonic data or a precomputed model/means series, but is not recommended for noisy visual data.

MMRD Relations

The default relation is:

$$M_{V,0} = 2.16 \log_{10}(t_2) - 10.804$$

from Kantharia (2017). It is the preferred modern single-relation calibration and remains first in the drop-down menu.

The plug-in also includes historical relations and two aggregate options:

- **Unweighted mean of historical relations (Kok 2010):** mean of the historical Kok relations; error is the inter-relation scatter. This is the best option for reproducing Kok's Table 3 and Table 4.
- **Inverse-variance weighted mean of historical relations:** weighted mean using the published coefficient errors and optional t_2/t_3 errors. Its reported error is the larger of the formal weighted error and the inter-relation scatter, avoiding over-precise error bars when calibrations disagree.

Individual historical relations are also available for comparison and study. After the exponential fit (or direct extraction) has supplied t_2 and t_3 , you can switch among these relations from the drop-down without starting the plug-in again. The results dialog also lists every relation in one table, so a single **OK** is enough to see how the calibrations differ.

Table 4 in Kok (2010)

Kok's Table 4 columns are not different MMRD relations. They are different extinction assumptions applied to the same Table 3 absolute magnitude:

- $A_V(1)$ and $D(1)$ use the Schlegel et al. (1998) full line-of-sight extinction.
- $A_V(2)$ and $D(2)$ use the Arenou et al. (1992) three-dimensional extinction model.
- $D(r)$ lists distances from other published work.

To reproduce Table 4, use Kok's Table 1 values for m_0 , t_2 , and t_3 , select **Unweighted mean of historical relations (Kok 2010)**, and enter the appropriate A_V value from Table 4.

Extinction

You can enter either:

- A_V directly, or
- reddening $E(B - V)$, in which case the plug-in uses:

$$A_V = 3.1E(B - V)$$

If both are supplied and A_V is non-zero, A_V takes precedence.

Error Bars and Bounds

The papers use error bars at several stages, and these do not all mean the same thing. The plug-in keeps the stages explicit:

Stage	Meaning
$\sigma t_2, \sigma t_3$	uncertainty in the decline times
σM_V for a single relation	propagated from published coefficient errors and optional t_2/t_3 errors
σM_V for unweighted mean	scatter among the historical relations
σM_V for inverse-variance mean	max(formal weighted error, inter-relation scatter)
distance bounds	σM_V propagated through the distance modulus

When the **Exponential model fit (Kok 2010, eq. 10)** source is used, the σt_2 and σt_3 fields are filled in automatically by propagating the fit's parameter uncertainties through the closed-form crossing time (see the Nova Exponential Decline Model documentation). If the fit cannot estimate those uncertainties, or the decline times are taken directly from the observations, the fields are left blank; you may enter values manually, for example from a published table.

Distance bounds are shown in kpc as lower and upper bounds rather than a single symmetric error, because distance depends exponentially on magnitude. A symmetric σM_V usually becomes an asymmetric

distance interval.

The results dialog repeats those bounds for the relation selected in the drop-down, and the summary table reports M_V , σM_V , and the corresponding distance interval for every relation, rounded to two decimal places. The table opens sorted by increasing distance; click any column header to sort by that column (numeric columns sort by value, not as text). Click a row to highlight it; the relation chosen in the drop-down is selected initially. Relations that cannot be evaluated (for example a t_3 calibration when t_3 is missing) appear as `n/a` and sort last.

Comparison with Kok (2010)

For each nova:

1. Match the JD window shown in Kok's figure.
2. Compare the fitted t_2 and t_3 with Table 1.
3. Compare M_V with Table 3.
4. Enter Kok's $A_V(1)$ or $A_V(2)$ and compare distance with Table 4.

For V4633 Sgr, using Kok's Table 1 values:

- $m_0 = 7.6$
- $t_2 = 19.5$
- $t_3 = 42.9$

the unweighted historical mean gives $M_V \approx -7.72$, close to Kok's published $M_V = -7.7 \pm 0.3$.

Limitations

- The plug-in does not look up extinction automatically.
- The Kok aggregate is a reconstruction; Kok's paper does not include source code or a fully specified weighting implementation.
- The MMRD method is empirical and calibration-dependent.
- Raw visual data can make peak detection and direct crossing detection fragile; use the exponential fit and inspect the model series.